

Ch16 Timer PWM Quiz ANS

1. What is the primary disadvantage of using Polling instead of Interrupts?

- A. Polling is more complex to implement.
- B. Polling wastes CPU cycles by continuously checking for events.
- C. Polling cannot handle multiple devices.
- D. Polling requires more hardware wiring than interrupts.

ANS: B

2. In the ARM Cortex-M architecture, what is the role of the Nested Vectored Interrupt Controller (NVIC)?

- A. To generate clock signals for the CPU.
- B. To coordinate multiple interrupt sources, manage priorities, and enable/disable interrupts.
- C. To automatically save all General Purpose Registers (R0-R15) to flash memory.
- D. To convert analog signals to digital signals for processing.

ANS: B

3. Where is the Interrupt Vector Table typically located in memory (before any relocation)?

- A. 0x20000000
- B. 0x08000000
- C. 0x00000000
- D. 0xFFFFFFFF

ANS: C (0x00000000 is the typical start address)

4. Which specific set of registers is automatically pushed onto the stack (Stacking) when an interrupt occurs?

- A. R0–R12, SP, LR, PC
- B. R0–R3, R12, LR, PC, xPSR
- C. R4–R11, SP
- D. All CPU registers (R0–R15)

ANS: B

5. During the Stacking process, the Stack Pointer (SP) is:

- A. Incremented by 32 bytes.
- B. Decrement by 32 bytes (for a full descending stack).
- C. Incremented by 4 bytes.
- D. Left unchanged; registers are saved to a special buffer.

ANS: B (Decrement by 32 bytes for 8 registers x 4 bytes)

6. What value does the Link Register (LR) hold when the processor is executing an Interrupt Service Routine (ISR)?

- A. The address of the next instruction in the main program.
- B. A special EXC_RETURN value (e.g., 0xFFFFFFFF9) indicating how to return.
- C. 0x00000000
- D. The address of the ISR itself.

ANS: B (EXC_RETURN value)

7. Which Stack Pointer is selected by default at Reset?

- A. Process Stack Pointer (PSP)
- B. Main Stack Pointer (MSP)
- C. Floating Point Stack Pointer
- D. System Stack Pointer

ANS: B (MSP is selected at reset)

8. If SPSEL bit in the Control Register is 0, which stack pointer is used in Thread Mode?

- A. MSP
- B. PSP
- C. Both MSP and PSP simultaneously
- D. None of the above

ANS: A (MSP is used if SPSEL = 0)

9. To exit an ISR and return to the main program, which instruction is typically used?

- A. MOV PC, #0
- B. BX LR
- C. POP {R4-R11}
- D. RET

ANS: B (BX LR triggers the unstacking process)

10. Which register is used to divide the CPU clock frequency to generate a slower Timer Clock frequency?

- A. Auto-Reload Register (ARR)
- B. Capture Compare Register (CCR)
- C. Prescaler Register (PSC)
- D. Control Register (CR1)

ANS: C (Prescaler Register)

11. If the CPU Clock is 16 MHz and PSC is set to 15999, what is the Timer Clock frequency (fCK_CNT)?

- A. 16 MHz
- B. 1 MHz
- C. 1 kHz
- D. 100 Hz

ANS: C ($16,000,000 / (15999 + 1) = 1,000 \text{ Hz} = 1 \text{ kHz}$)

12. In "Up-counting Mode," what happens when the Counter reaches the value in the Auto-Reload Register (ARR)?

- A. It stops counting immediately.
- B. It resets to 0 and generates an Update Event (UEV).
- C. It reverses direction and starts counting down.
- D. It waits for a software reset trigger.

ANS: B (Resets to 0 and generates UEV)

13. The formula for the Timer Period in Up-counting mode is:

- A. $(ARR + 1) * \text{Clock Period}$
- B. $ARR * \text{Clock Period}$
- C. $(PSC + 1) * ARR$
- D. $2 * ARR * \text{Clock Period}$

ANS: A

14. In "Center-aligned Counting Mode," the counter counts:

- A. Up from 0 to ARR, then resets to 0.
- B. Down from ARR to 0, then resets to ARR.
- C. Up from 0 to ARR, then Down from ARR to 0.
- D. Randomly between 0 and ARR.

ANS: C

15. Which PWM Mode behaves as "Low True" (Active Low) in Up-counting mode?

- A. PWM Mode 1
- B. PWM Mode 2

C. Center-aligned Mode

D. Capture Mode

ANS: A (Mode 1 is Low True/Active when $CNT < CCR$)

16. In PWM Mode 1 (Up-counting), if $CNT < CCR$, the output is:

A. Active (High)

B. Inactive (Low)

C. Toggled

D. Undefined

ANS: A

17. To increase the Duty Cycle in PWM Mode 1, you should:

A. Decrease the CCR value.

B. Increase the CCR value.

C. Set ARR to 0.

D. Increase the Prescaler (PSC).

ANS: B (Higher CCR means the "Active" condition $CNT < CCR$ lasts longer)

18. How is the Duty Cycle calculated for PWM Mode 1?

A. Duty Cycle = $1 - (CCR / (ARR + 1))$

B. Duty Cycle = $CCR / (ARR + 1)$

C. Duty Cycle = $(ARR + 1) / CCR$

D. Duty Cycle = PSC / ARR

ANS: B

19. A timer runs from a 72 MHz clock. $PSC = 71$ and $ARR = 999$. In PWM Mode 1, $CCR = 250$. What is the duty cycle?

A. 10%

B. 25%

C. 50%

D. 75%

ANS: B (Duty = $CCR / (ARR + 1) = 250 / 1000 = 25\%$)

20. In center-aligned mode, a timer has $ARR = 4000$, PSC chosen so that the counter clock is 50 kHz. What is the PWM period?

A. 40 ms

B. 160 ms

C. 320 ms

D. 640 ms

ANS: B (Period = $2 * ARR * T_{clk} = 2 * 4000 / 50,000 = 0.16 \text{ s} = 160 \text{ ms}$)